Jonathan G. Barata

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Education

B.S. in Electrical Engineering — ***California Institute of Technology (Caltech)***
Track: Intelligent Systems — *Pasadena, California*
Year: Junior — *Sep. 2024 - Present*

Projects

Wallis – Wheeled Legged Balancing Robot — *June 2026 - Sep. 2026*
Independent Robotics Project — *Pasadena, California*

- Designed the full electromechanical stack for a 1 m tall, 8 kg robot capable of balancing with a variety of controllers.
- Tested in MuJoCo simulation to verify full squat-to-stand motion and recovery from disturbances.
- Designed a custom 4 layer high-voltage PCB to integrate motor communication and control circuits.

Autonomous Mapping & Navigation Robot — *Mar. 2026 - May 2026*
Caltech ME/EE 129 – 2 Person Team Project — *Pasadena, California*

- Integrated IR line following with gyro estimates and ultrasonic sensors to develop an exploration system that autonomously discovers roads, intersections, blockages, and dead ends on a map.
- Implemented search algorithms for shortest-path routing and directed exploration.

Stereo Vision Ball Tracking & Landing Prediction — *Jan. 2026 - Mar. 2026*
Caltech CS 12 – Computer Vision Project — *Pasadena, California*

- Designed and built a dual-camera system capable of tracking the 3D trajectory of a ball.
- Used the calculated trajectory with a Kalman filter to predict landing spots on the ground with 5 cm accuracy 0.25–0.5 s before impact.

Work Experience

Research Intern — *Sep. 2026 - Present*
Autonomous Robotics and Control Lab at Caltech — *Pasadena, California*

- Created a testing sequence for ROS 2 communication with a Booster T2 humanoid robot.
- Trained and deployed a locomotion and box pick-up policy.

Head Altium Teaching Assistant — *Sep. 2025 - Dec. 2025*
EE13: Electronic Systems Prototyping – Caltech — *Pasadena, California*

- **PCB Design Instruction:** Appointed Head Altium TA and led instruction on schematic capture, PCB layout, and manufacturing-ready board design in Altium.
- **Technical Mentorship:** Mentored more than 30 students through design reviews, hardware debugging, embedded systems development, and PCB prototyping.

Technical Skills

Languages	Python, C++, Git, Matlab
Robotics & Simulation	ROS 2, Isaac Sim, MuJoCo, OpenCV, URDF
Controls, Planning & ML	Reinforcement Learning, LQR, MPC, PID, Kalman Filtering, Dijkstra, A*
Hardware & Design	Altium, PCB Fabrication & Assembly (JLCPCB), Fusion 360, CAN, Teensy, Arduino, Raspberry Pi, 3D Printing